

LESSON GUIDE

Technology Plus

Unit 2: Programming

1.2.1 – Binary

Lesson Overview:

In this lesson, students will explore how computers use the binary number system to represent and process data, practice converting between binary and decimal, and apply their knowledge by encoding and decoding binary messages in a hands-on networking activity.

Students will:

- Understand the importance of binary in computing.
- Convert decimal numbers to binary in the CYBER.ORG Range using CyberChef.

Guiding Question: Why do computers only speak in 1s and 0s, and how can we learn their language?

Suggested Grade Levels: 8-12

Technology Needed: Yes

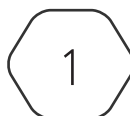
Approximate Time Required: 55-65 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 1.2 – Identify notational systems.
 - Binary
 - Hexadecimal
 - Decimal
 - Octal

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Lesson Background

Background Information:

Computing is built on binary, but most users never think about how data is actually stored, transmitted, or processed at its lowest level. Without a foundational understanding of binary, students may struggle later when tackling networking, programming, or cybersecurity concepts that rely on binary logic.

This lesson gives them a hands-on way to engage with binary beyond theory – seeing how computers interpret numbers, text, and even messages in real-time. By the end, students aren't just memorizing conversions; they're building an intuition for how digital systems work under the hood.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 1.2.1 – Binary • Lab Slides 1.2.1 – Conversions with CyberChef
Materials per Student	• Student Notes 1.2.1 – Binary • Student Handout 1.2.1 – Binary • Assessment 1.2.1 – Binary (optional)

Cyber Terms:

Term	Definition
binary	a base-2 number system that only uses two digits: 0 and 1, commonly used to represent digital data in computing
decimal	a base-10 number system that uses the digits: 0 through 9; the standard system humans use for counting and arithmetic
bit (b)	short for "binary digit", represented as a 0 or 1 and comprises the most basic unit of information storage in computing
byte (B)	collection or unit of eight bits, sometimes subdivided into two halves referred to as a "nibble"

Lesson Background

Cyber Terms:

Term	Definition
logic gates	the building blocks of digital circuits that use binary values to perform operations like AND, OR, and NOT, controlling how computers process information
American Standard Code for Information Interchange (ASCII)	a character encoding system that assigns a unique number (binary code) to letters, numbers, and symbols so computers can store and process text
Unicode	an expanded character coding system that includes characters from multiple languages, special symbols, and emojis, making it more versatile than ASCII

1.2.1 – Binary

Guiding Question: Why do computers only speak in 1s and 0s, and how can we learn their language?

The Bits on Binary (20-25 minutes)

- Distribute Student Handout 1.2.1 – Binary and encourage students to complete the handout as you go through today's lesson.
- The guiding question on slide two will also be in the lab slide you provide to students so that they can copy and paste the binary into CyberChef.
- In mathematics and computer science, different number systems are often notated using subscripts to indicate their base. This helps avoid confusion when working with multiple systems.
- This lesson only uses binary and decimal, but moving forward, hexadecimal and octal will be added, so introducing this concept to your students could be helpful.
- **Decimal** (base-10) is usually normally written but can be explicitly noted as 123_{10} .
- **Binary** (base-2) is written with the subscript 2, like 1011_2 , to indicate it's a binary number.
- **Hexadecimal** (base-16) uses a 16 subscript, like $2F_{16}$.
- **Octal** uses an 8 subscript like 75_8 .
- In programming, subscripts aren't used, so other notations exist:
 - Binary: often written as 0b1011
 - Hexadecimal: written as 0x2F
 - Octal: written as 075 in some older programming languages
- When converting to binary from decimal, some students may need a refresher on dividing to just get remainders!
- Students may be confused as to why 8 bits stops at 255 instead of 256 (forgetting that 0 is an option). You could draw a place value chart with the corresponding powers of 2 on the board. Students can add these place values together and confirm they equal 255.

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1

- This lesson briefly touches on storage units. Lesson 1.3.1 Storage Units in Unit 3 covers the topic in much more detail.
- The numbers on slide 9 are listed in the Student Handout for students to record their answers. Answers are animated to appear on the slide when you are ready to review them with students.
- ASCII and Unicode were specifically listed as topics to be aware of in FC0-U61, but not in FC0-U71.

They remain in this lesson since ASCII relates specifically to how the characters we create when we type are recognized and stored on the computer.

Lab: Conversions with CyberChef (30 minutes)

- Numeric conversions can be done on a Windows computer using the regular calculator that is installed. The calculator can be switched from Standard to Programmer and show students conversions to binary, decimal, hexadecimal, and octal.
- Inform students that capitalization and spelling count when using the terminal. If students are having problems, have them check how they are typing.
- The CyberChef parts of the lab should be pretty easy. Students will either use the Recipes “From Decimal, To Binary”; “From Binary, To Decimal”; or just “To Binary.”
- **CyberChef**, **send_binary1**, and **send_binary2** are command aliases that are specific to the CYBER.ORG Range Kali machine only. The **CyberChef** just uses **xdg-open** to open Firefox directly to CyberChef. The **send_binary1** and **send_binary2** aliases establish network listeners on port 6666 using Netcat so that students can message back and forth. The aim in the naming of the aliases was that students would associate the command with this lesson and not realize they can use it going forward just to chat with other students.
- The Putting It All Together questions on slide 19 are on the student handout for students to record their answers.
- As an interesting final wrap to this lab, ask the students if they realize they were turning messages into 1s and 0s that the computer then had to turn into 1s and 0s to send to their partner to then turn into 1s and 0s that the partner could decode!

Putting It All Together (5-10 minutes)

- These Putting It All Together questions are not on the handout. You can discuss them or create an Exit Ticket with them if you like.

- What was today's guiding question?

Why do computers only speak in 1s and 0s, and how can we learn their language?

- How would you answer today's guiding question based on what you have learned?

Computers speak in binary. It is a base-2 number system that computers use because their hardware is built on electronic circuits that only recognize two states, on (1) and off (0). We can learn to speak their language by converting binary to decimal or converting binary to ASCII.

Decimal is the number system we are most used to counting with and ASCII converts letters, characters, symbols into binary for storage on computers.

- What is a bit and how does it relate to a byte?

A bit, short for binary digit, is the smallest unit (either a 0 or a 1), and 8 bits make up a byte.

- How do you convert binary to decimal?

Converting binary to decimal involves multiplying the 0 or the 1 by the power of 2 its placement represents and then adding those values together to get decimal result.

- Understanding binary is essential for anyone working with technology, as it forms the foundation of all digital systems!